Experiment 5: Word Stemming using Porter Stemmer in Python

1. Aim

To write a Python program to use the Porter Stemmer algorithm to perform word stemming on a list of words.

2. Procedure

1. Import `PorterStemmer` from the `nltk.stem` module.
2. Create an instance of the `PorterStemmer` class.
3. Define a list of words with different morphological suffixes.
4. Apply `ps.stem()` to reduce each word to its root stem.
5. Display the original words and their stemmed outputs.

3. Program

```
from nltk.stem import PorterStemmer

ps = PorterStemmer()
words = ["program", "programs", "programmer", "programming", "programmers"]

stemmed = [ps.stem(w) for w in words]
print("Original Words:", words)
print("Stemmed Words:", stemmed)
```

4. Output

```
Original Words: ['program', 'programs', 'programmer', 'programming',
'programmers']
Stemmed Words: ['program', 'program', 'programm', 'program', 'programm']
```

5. Result

Thus, the Python program was successfully executed to perform word stemming on a list of words using `PorterStemmer`.